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Technical English Vocabulary In Use

Digital Signal Processing:

Playback speed variation:

- Popularized definition:

The playback speed variation refers to the adjustment of the speed at which audio is played back, making it faster or slower than its original recording speed

- Technical definition:

The playback speed variation refers to the alteration of the rate at which audio content is played, either speeding it up or slowing it down compared to its original version. This can affect the pitch and timing of audio, depending on the degree of variation.

Noise:

- Popularized definition:

Noises are unwanted, random and unexpected sounds or signals that mix with the main signal that we are trying to process, making it harder to get a clear result.

- Technical definition:

Noises refers to any unwanted or random disturbance that interferes with the desired signal. It can originate from various sources such as thermal, quantization, or electromagnetic interference. Noise is often modeled and filtered out to enhance the quality of the signal.

Hertz:

- Popularized definition:

Hertz is the unit used to measure how often something happens in a second

- Technical definition:

Hertz is the unit of frequency that measures the number of cycles per second of a periodic signal.

Fingerprint:

- Popularized definition:

Fingerprint refers to a small and unique summary of a signal that helps identify it by focusing on its important features.

- Technical definition:

Fingerprint refers to a compact and unique representation of a signal's key features, generally used for identification or comparison. This representation is derived through feature extraction techniques such as Fourier transforms.

Spectrogram:

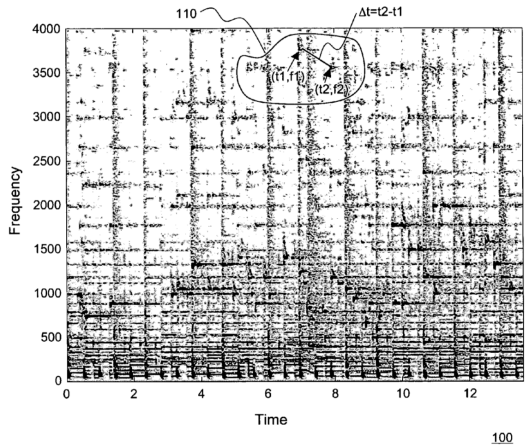
- Popularized definition:

A spectrogram is like a colorful diagram that shows how different signals change over time. It displays the frequencies and their strengths, to provide insights into sound patterns

- Technical definition:

A spectrogram is a visual representation of the spectrum of frequencies in a signal over time. It displays frequency, amplitude, and time information, making it possible to analyze the signal's characteristics and changes in detail

Example of a spectrogram:



Distortions:

- Popularized definition:

Distortion is when a signal gets changed or messed up during processing, making the output look or sound different from the original input

- Technical definition:

Distortion happens when the original signal's waveform gets altered during processing. This can occur when the output signal differs from the input due to various factors, such as non-linearities and quantization errors. As a result, there might be changes in amplitude or frequency components.

Audio sample:

- Popularized definition:

An audio sample is a tiny snapshot of sound taken at a specific moment, which helps turn a continuous sound wave into digital data.

- Technical definition:

An audio sample is a specific measurement of an analog audio signal's amplitude at a particular moment in time. These samples are usually captured at regular intervals based on the sampling rate. They help convert the continuous analog signal into a digital format.

Magnitude:

- Popularized definition:

Magnitude on a signal graph represents the strength of a signal at some points. It helps to see how the signal's intensity varies over time to understand how we can hear the different levels of sounds

- Technical definition:

Magnitude in signal processing graphs indicates the strength of a signal at specific frequencies. It shows how the signal's power changes with frequency and is essential for understanding system behavior and response

Applied Mathematics:

Shear transformation:

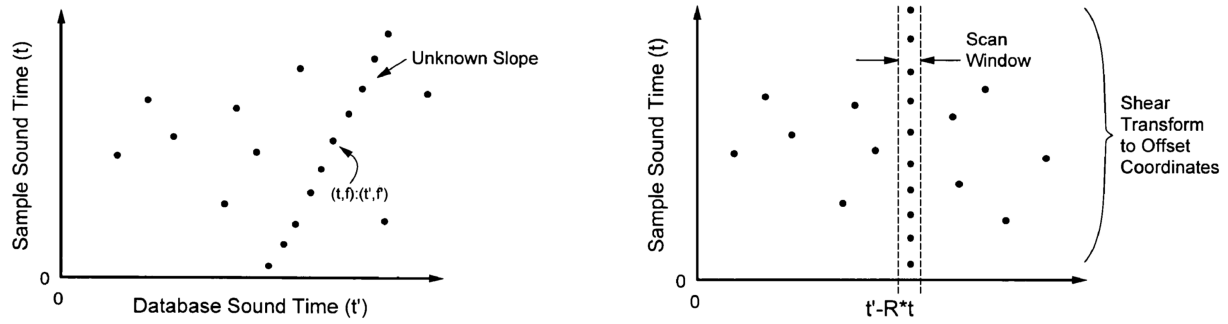
- Popularized definition:

A shear transformation changes the shape of an object by sliding its sides in a parallel direction while keeping one side fixed, making the object appear stretched

- Technical definition:

A shear transformation alters the shape of an object by shifting its points in a direction parallel to one axis while keeping the other coordinate constant, making the object appear stretched

Example of a shear transformation:



A linear regression is a method used to understand how two elements are connected by drawing a straight line through data points on a graph. It allows us to see trends and make predictions based on the data we have collected.

- Technical definition:

A linear regression is a statistical technique that models the relationship between a dependent variable and other independent variables by fitting a straight line to the data. It is used to quantify the strength of the relationship and make predictions.

Outliers:

- Popularized definition:

Outliers are unusual data points that don't fit in with the other. They might be much higher or lower than the other.

- Technical definition:

Outliers are data points that differ significantly from the rest of the dataset. They can indicate variability or measurement errors.

Scatter-plot:

- Popularized definition:

A scatter plot is a type of graph that shows how two things are related by marking points on a grid based on their values.

- Technical definition:

A scatter plot is a graphical representation of data points on a two-dimensional plane, where each point represents the values of two variables. The horizontal axis (x-axis) and vertical axis (y-axis) are used to plot the corresponding values of each data point.

Clump:

- Popularized definition:

A group of elements close to each other, like trees in a forest. They are studied by experts to find patterns or connections, which helps in understanding how data or objects are grouped.

- Technical definition:

A dense region in a dataset where elements are highly similar, generally identified through clustering or density-based techniques. It shows us non-uniform distributions or correlations

Equation:

- Popularized definition:

An equation is a math sentence with an equal sign that shows two things are the same.

- Technical definition:

An equation is a mathematical statement that asserts the equality of two expressions, typically involving variables and constants, separated by an equal sign (=). It expresses a relationship where one side is equal to the other under certain conditions.

Continuous function:

- Popularized definition:

A function is continuous if you can draw its graph without lifting your pen from the paper. In other words, there are no breaks, jumps, or holes in the graph of the function.

- Technical definition:

A continuous function can be formally defined as a function $f : X \to Y$ where the pre-image of every open set in Y is open in X. More concretely, a function f(X) in a single variable x is said to be continuous at point x_0 if

1. $f(x_0)$ is defined, so that x_0 is in the domain of f.
2. $\lim x \to x_0 f(x)$ exists for x in the domain of f.
3. $\lim x \to x_0 f(x) = f(x_0)$

Neighbor:

- Popularized definition:

A neighbor of a point is any point that is close to it within a certain range, or the area around that point.

- Technical definition:

A neighbor refers to an element that is immediately adjacent to another element in a given structure or space. In topology, two points are neighbors if they belong to the same open set. In a sequence or array, neighbors are elements that are next to each other.

CHAT GPT:

Spectrogram: Imagine a heatmap or a colorful graph that shows how the sounds (or frequencies) in a piece of audio change over time. It helps visualize which frequencies are present at different moments, much like how a weather radar shows changing conditions.

Fingerprint Objects: These are unique "tags" or pieces of information that help identify specific parts of an audio sample, similar to how a fingerprint identifies a person. Each "tag" highlights a distinct feature of the sound.

Fourier Transform: This is like a special tool that breaks down a complex sound or signal into its basic building blocks (individual frequencies), making it easier to analyze or understand the different sounds within.

Frequency Bins: Think of these as buckets that collect different parts of sound. Each bin holds a specific range of frequencies (high or low-pitched sounds), and together, they help divide the audio into manageable pieces.

Histogram: A graph that counts how often something happens. In audio processing, it helps detect patterns, such as finding a repeated sound by counting how frequently certain values appear.
Invariant Component: This is the part of the fingerprint that stays the same no matter how the sound changes (like speeding up or slowing down), similar to how your fingerprints remain the same even if your hands grow.

Variant Component: Unlike the invariant component, this part can change depending on how the sound is played. For example, if you speed up a song, the pitch may go higher, and this component tracks such changes.

Relative Value: A comparison between two similar parts of different sounds. It’s like measuring how much one version of a song is sped up compared to another version.

Time-Frequency Analysis: A method of looking at both when and how often certain sounds happen within a piece of audio. It's like reading a music sheet that shows not only the notes (pitches) but also when each note is played.

Wigner-Ville Distribution: A complex tool that gives a very detailed picture of how sound frequencies change over time, almost like a highly sensitive radar that tracks every small movement of a storm.

Wavelets: These are tiny, short waves that can be used to analyze a sound in fine detail. It’s like zooming in on a sound to see its smaller parts, much like using a magnifying glass on a picture to reveal more details.

Cepstral Coefficients: These are special measurements used in audio recognition, like how voice assistants recognize your voice. They capture the "shape" of the sound, helping a system understand speech or music.

Peak Detection: This refers to finding the highest points or the most intense parts of a sound, much like finding the loudest notes in a song.

Super-Resolution Techniques: These are methods that make the details of a sound clearer than they normally would be, like zooming into a blurry photo and making it sharp again.

Parabolic Interpolation: This is a way to get a more precise estimate of sound frequencies by filling in the gaps, like connecting dots smoothly to get a more accurate picture of a curve.

Time Stretch: A process where a sound is slowed down or sped up without changing its pitch. Imagine listening to a song in slow motion but the notes still sound right—just stretched out in time.

Difference between our work and ChatGPT's:
Our work is similar to ChatGPT's, as we have both provided popularized definitions. However, our definitions are more precise, while those of ChatGPT may seem vaguer and less detailed. This means that ChatGPT's definitions are easier to understand but they look like they are made for children, they have no technical parts even if it is very easy to understand, ours offer additional clarity and precision that can help better grasp the concepts.
We haven’t seen any error on the text it generated.

Description of graphs:

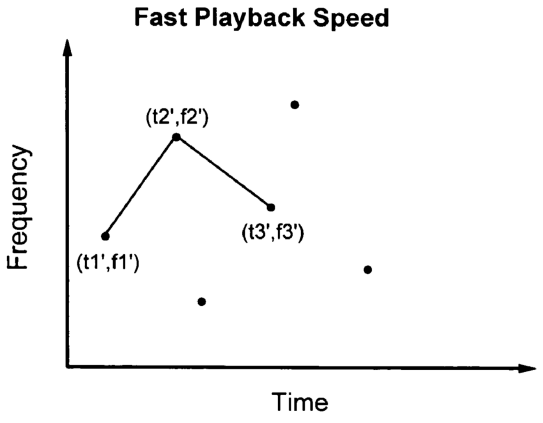


FIG. 5B

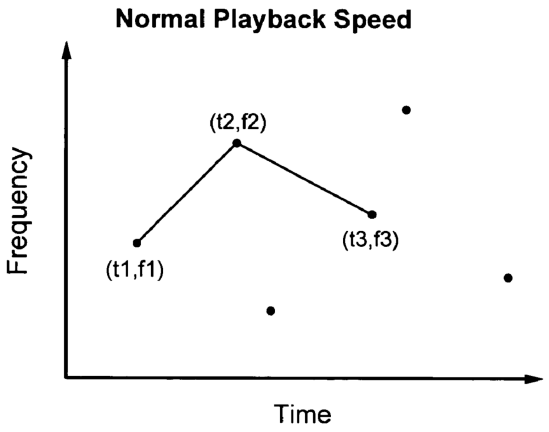


FIG. 5A

This graph (FIG 5B) displays a scatter plot with time on the x axis and frequencies on the y axis. Each point on the graph represents a specific frequency at a given time. While some points are scattered across the plot, some other of them are connected. The graph shows a few spectral peaks with their respective coordinates labeled as (t1′,f1′), (t2', f2'), and (t3', f3'), where "t" represents time and "f" represents frequency. These points represent significant frequency values in an audio signal at specific time intervals.

If we compare this graph with the “normal” version of it (FIG 5A), we can see that the points are more spaced out on the x axis due to the speed of the time. The y axis hasn't changed because the frequency dont change.